Answer key · for the teacher

Sensors — answers

IoT

Answers for the four pupil worksheets and the unit check. Where pupils write their own ideas, a sample answer shows the kind of thinking to look for — accept any answer that shows the same idea.

Lesson 1 worksheet: Sensor spotter

What is a sensor? · Find the parts that notice, show and act

1 Sense or act?

a. Thermometer on the classroom wall — **Senses (sensor)**
b. Pump that waters the garden — **Acts (actuator)**
c. Microphone in a phone — **Senses (sensor)**
d. Buzzer that sounds an alarm — **Acts (actuator)**
e. Light sensor in a solar lamp — **Senses (sensor)**
f. Motor that turns a fan — **Acts (actuator)**
g. Probe that checks soil moisture — **Senses (sensor)**
h. Loudspeaker playing music — **Acts (actuator)**

2 What does it sense?

1. Thermometer → **How hot or cold it is**
2. Soil probe → **How wet the soil is**
3. Microphone → **Sound**
4. Light sensor → **How bright or dark it is**

3 My sensor

Sample answer: A drawing of the solar lamp at the school gate. “Its light sensor notices when it gets dark, and then the lamp switches on.”

☆ Challenge

Sample answer: Sensor: a soil-moisture probe in the garden bed. Actuator: a pump or a tap that lets water flow. Deciding: a small computer with a rule like “dry → water”, or a pupil reading the probe.

Lesson 2 worksheet: Reading detective

Reading numbers: units and scales · Number, unit, time, place — and read it level

1 True or false?

a. “26” on its own is a complete reading. — **False**
b. 30 °C and 30 °F are the same temperature. — **False**
c. Read a scale with your eye level with the mark. — **True**
d. A thermometer in direct sun can read higher than one in shade. — **True**
e. Readings at 7 am and 2 pm make a fair comparison of two days. — **False**

2 Match the units

1. 28 °C → **How hot or cold it is**
2. 15 mm → **How much rain fell**
3. 40 % → **40 out of every 100**
4. 8 am, school gate → **When and where it was read**

③ Think it through

a. A scale has 5 equal steps from 20 to 30. What is each step worth? — **2 °C, because 10 ÷ 5 = 2.**

b. Why can't you compare a 7 am shade reading with a 2 pm sun reading? — **The time and the sun both change the reading, so the difference may not come from the weather.**

c. Rewrite "Rain: 12" as a full reading. — **For example: 12 mm, 8 am Monday to 8 am Tuesday, school field.**

☆ Challenge

Sample answer: Use the same thermometer, read at the same time (10 am), wait five minutes in each spot, read at eye level and write °C each time. Only the place changes.

Lesson 3 worksheet: Rain gauge builder

Build a rain gauge · Build it, place it, read it in millimetres

① Build it in order

1. An adult cuts the top off the bottle → **2.** Add stones and water; mark the 0 line →
3. Tape a millimetre scale up from 0 → **4.** Put the top in upside down as a funnel →
5. Place it in open ground

② Good spot or bad?

a. Middle of the open school field — **Good spot**

b. Under the big tree by the gate — **Bad spot**

c. Under the edge of the classroom roof — **Bad spot**

d. Open garden, far from walls and trees — **Good spot**

e. Inside the classroom by the window — **Bad spot**

f. Open ground, held firm with stones — **Good spot**

③ Read the gauge

a. Why must the bottle have straight sides? — **So every millimetre on the scale matches a millimetre of rain.**

b. The water is 14 mm above 0. What do you write in the log? — **14 mm, with the date and the time, for example 8 am.**

c. Why do we pour it back to 0 after reading? — **So no rain is counted twice: tomorrow's reading shows only new rain.**

☆ Challenge

Sample answer: The tree's leaves caught some of the rain before it reached Zainab's gauge, so her reading is too low. Musa's gauge was in the open.

Lesson 4 worksheet: Rain data detective

Data in class: record, chart, decide · Record, chart, spot the pattern, decide

① Steps in order

1. Record readings in a table → **2.** Draw a bar chart → **3.** Spot the pattern → **4.** Decide what to do

② Draw the chart

Sample answer: Five bars with “Rain (mm)” up the side in equal steps: Tue tallest (10), then Mon (6) and Wed (4); no bars for Thu and Fri. Pattern: two dry days at the end.

③ True or false?

a. Write 0 when nobody read the gauge. — **False**

b. Each bar’s height shows the rain in mm. — **True**

c. The steps up the side must be equal. — **True**

d. In task 2, Thursday and Friday were dry. — **True**

e. A connected sensor never needs checking. — **False**

☆ Challenge

Sample answer: Yes. Thursday and Friday both had 0 mm — less than 5 mm in two days — so the rule says water. Feel the soil first to check.

Unit check — answers and reasons

10 questions · the app gives this feedback after each answer

1. **B — The light sensor that switches on a solar lamp at dusk.** It notices the daylight fading and turns that into a signal that switches the lamp on.
2. **Sensor → Notices something and turns it into a number; Display → Shows the reading to people; Actuator → Acts, such as pumping water or sounding a buzzer.** Sense, show, act. In between, a person or a small computer decides what to do with the reading.
3. **False.** It needs a unit (°C), plus a time and a place, so others can check it and compare it fairly.
4. **A, B, D. A — yes:** The same time of day removes the morning-to-afternoon change. **B — yes:** Sun heats the thermometer itself, so keep the place the same. **C — no:** The sun alone can push one reading up, so you can’t tell which day was warmer. **D — yes:** Reading at an angle shifts the number, so read both the same careful way. **E — no:** Mixed units make the numbers look different even when the temperature is the same.
5. **1. An adult cuts off the bottle’s top → 2. Add stones and water; mark the 0 line → 3. Add a mm scale and the funnel → 4. Place it in open ground → 5. Read it at the same time each day.** Build it, place it, then read it at the same time daily — and pour it back to 0 after each reading.
6. **A — So every millimetre on the scale matches a millimetre of rain.** Where a bottle narrows, a little rain rises a long way up the scale and reads too high.
7. **open, time, zero.** Open ground keeps roofs and trees from changing the catch; the same time makes each reading cover one full day.
8. **B — Ask pupils to redraw the side in equal steps, then compare the two charts.** Pupils see how unequal steps bend the picture — a lesson they’ll remember. The wall chart now tells the truth at a glance.
9. **A, B. A — yes:** It can take a reading every few minutes and send it over a network. **B — yes:** That’s the decide step, done automatically. **C — no:** A badly placed or faulty probe gives wrong data. People still check. **D — no:** Connected systems need power and a network. Your board chart needs neither.
10. **False.** A blank could mean no rain or no reading. Write 0 for dry days and “missed” when nobody read the gauge.